Health Sciences Tech - Class 9 (PCTB)

Class 9 — Solved Exercises

Punjab Curriculum and Textbook Board (PCTB)

8 chapters — 202 exercises

Chapter 1: Introduction to Health Sciences

A. Select the correct answers for the following questions.

1. A person feels weak, but medical tests are normal. This is:

(a) A disease

(b) A disorder

(c) An illness ✓

(d) An infection

Correct answer: An illness

A. Select the correct answers for the following questions.

2. What is the best definition of "health"?

(a) No illness or disease.

(b) A fit and active body.

(c) Physical, mental, and social well-being. ✓

(d) Ability to do daily tasks.

Correct answer: Physical, mental, and social well-being.

A. Select the correct answers for the following questions.

3. A person exercises but feels lonely. What is affected?

(a) Physical health

(b) Mental health

(c) Social health ✓

(d) Emotional health

Correct answer: Social health

A. Select the correct answers for the following questions.

4. Lack of clean water mostly can affect:

(a) Physical health ✓

(b) Social health

(c) Mental health

(d) Emotional health

Correct answer: Physical health

A. Select the correct answers for the following questions.

5. Which disease spreads from person to person?

(a) Diabetes

(b) Influenza

(c) Hypertension

(d) Cancer

Correct answer: Influenza

A. Select the correct answers for the following questions.

6. Which statement is correct for non-communicable diseases?

(a) Cannot spread from person to person

(b) Caused by bacteria and viruses

(c) Treated with antibiotics

(d) Spread through air and water

Correct answer: Cannot spread from person to person

A. Select the correct answers for the following questions.

7. Which of these is a communicable disease?

(a) Asthma

(b) Stroke

(c) Tuberculosis

(d) Arthritis

Correct answer: Tuberculosis

A. Select the correct answers for the following questions.

8. Which method best controls non-communicable diseases?

(a) Healthy lifestyle and early screening ✓

(b) Taking antibiotics

(c) Wearing masks

(d) Avoiding public places

Correct answer: Healthy lifestyle and early screening

A. Select the correct answers for the following questions.

9. Which institutions provide primary healthcare in Punjab?

(a) District Headquarters (DHQ) hospitals

(b) Tehsil Headquarters (THQ) hospitals

(c) Basic Health Units (BHUs) ✓

(d) Teaching Hospitals

Correct answer: Basic Health Units (BHUs)

A. Select the correct answers for the following questions.

10. Which healthcare institution is most suitable for heart surgery?

(a) Tehsil Headquarters (THQ) hospital

(b) Teaching hospital ✓

(c) Rural Health Center (RHC)

(d) Basic Health Unit (BHU)

Correct answer: Teaching hospital

B. Write short answers.

1. Differentiate between illness, disease, and disorder with examples.

Illness is the person's own feeling of being unwell, for example feeling weak even though medical tests are normal. Disease is a condition that can be found

and diagnosed objectively, for example diabetes shown by a blood-sugar test. Disorder is an abnormality of function or structure, for example an anxiety disorder, which disturbs normal mental functioning even without a single germ or injury.

B. Write short answers.

2. What are the three main dimensions of health? Describe each briefly.

Physical well-being means a fit body, free from disease, with energy for daily work. Mental well-being means balanced emotions, clear thinking and the ability to cope with stress. Social well-being means healthy relationships, a sense of belonging and support from family and community.

B. Write short answers.

3. Differentiate between communicable and non-communicable diseases.

Communicable diseases spread from person to person through germs, for example tuberculosis, dengue and hepatitis. Non-communicable diseases do not spread between people; they develop from lifestyle, genes or age over time, for example diabetes, heart disease and cancer, and are controlled mainly by a healthy lifestyle and early screening.

B. Write short answers.

4. Briefly describe the functions of three levels of healthcare in Punjab?

Primary care is the first contact: Basic Health Units (BHUs) and Rural Health Centres (RHCs) give basic treatment, prevention, vaccination and maternal care near people's homes. Secondary care is specialist hospital care at Tehsil Headquarters (THQ) and District Headquarters (DHQ) hospitals for patients referred from primary care. Tertiary care is advanced specialist care at teaching hospitals, including major surgery, for the most serious cases.

B. Write short answers.

5. Define Infant Mortality Rate (IMR) and Maternal Mortality Rate (MMR).

Infant Mortality Rate (IMR) is the number of infants who die before their first birthday, per 1,000 live births in a year. Maternal Mortality Rate (MMR) is the number of mothers who die from pregnancy-related causes, per 100,000 live births. Both are key indicators of a population's health and of its healthcare quality.

C. Write answers in detail.

1. Describe the three dimensions of health: physical, mental, and social well-being. How are they interrelated?

Physical well-being is bodily fitness and freedom from disease; mental well-being is emotional balance and the ability to handle stress and think clearly; social well-being is good relationships and community support. They are interrelated: a long physical illness can cause low mood and isolation, poor mental health can weaken the body and appetite, and loneliness or family stress can harm both mind and body. That is why the WHO definition treats health as all three together, not merely the absence of disease.

C. Write answers in detail.

2. Classify non-communicable diseases with examples.

Non-communicable diseases can be classified as: cardiovascular diseases, such as heart disease, hypertension and stroke; metabolic diseases, such as diabetes; cancers, such as lung or breast cancer; and chronic respiratory diseases, such as asthma and chronic obstructive pulmonary disease (COPD). They share risk factors like unhealthy diet, tobacco, alcohol, inactivity and obesity, and they are usually long-lasting, so prevention and early screening matter more than a single cure.

C. Write answers in detail.

3. Explain the structure of the healthcare system in Punjab.

Punjab's healthcare system has three levels. Primary care (BHUs and RHCs) is the community's first contact, giving basic treatment, immunization, maternal and child care and health education. Secondary care (THQ and DHQ hospitals) receives referred patients and provides specialist doctors, surgery and inpatient care at tehsil and district level. Tertiary care (teaching hospitals) provides highly specialized treatment, major surgery and medical training, and also handles the most complex referrals from the lower levels.

C. Write answers in detail.

4. Compare Basic Health Units (BHUs), Rural Health Centres (RHCs), and Tehsil Headquarters (THQ) hospitals.

A BHU is the smallest rural unit, staffed for basic outpatient care, immunization and maternal care for nearby villages. An RHC is larger, serving a wider rural area with more staff, beds and basic inpatient and laboratory services. A THQ hospital is a tehsil-level hospital with specialist doctors, an operating theatre and inpatient wards, receiving referrals that BHUs and RHCs cannot manage. In short, size, staff, equipment and the seriousness of cases handled all rise from BHU to RHC to THQ.

C. Write answers in detail.

5. What are the functions of District Headquarters (DHQ) hospitals and Teaching hospitals?

A DHQ hospital is the main hospital of a district: it provides specialist care, surgery, emergency and inpatient services, and it supervises and receives referrals from THQ hospitals, RHCs and BHUs in the district. A teaching hospital is attached to a medical college: besides advanced specialist care and major

surgery (for example heart surgery), it trains doctors, nurses and allied health students and carries out medical research, so it handles the most complex cases in the province.

Group Activities

1. Write a short essay on the importance of primary healthcare in Punjab. (Model answer)

Model essay: Primary healthcare matters most in Punjab because it reaches people where they live. BHUs and RHCs treat common illnesses early, vaccinate children, care for mothers, and teach hygiene, so small problems do not become serious hospital cases. It is also the cheapest level of care, which protects poor families from heavy medical costs. A strong primary system therefore improves health indicators like infant and maternal mortality, reduces pressure on DHQ and teaching hospitals, and keeps communities healthier overall.

Group Activities

2. Create a poster on the prevention and management of hepatitis in Punjab. (Model guide)

Model poster guide: Title: Protect Yourself from Hepatitis. Points: wash hands and drink clean water; get vaccinated against hepatitis; never share razors, needles or toothbrushes; ensure blood is screened before transfusion; eat hygienically prepared food. Management line: if you feel tired, lose appetite or notice yellow eyes/skin, visit a BHU or doctor at once and follow the prescribed treatment fully; do not self-medicate.

Group Activities

3. Write the challenges of managing cardiovascular diseases in Punjab. (Model answer)

Model answer: The main challenges are late diagnosis, because many people ignore high blood pressure or chest discomfort until a crisis; lifestyle risks, including salty and fatty diets, tobacco, inactivity and obesity; limited access to specialist cardiac care in rural areas; the high cost of long-term medicines and procedures; and low awareness of warning signs and of the need for regular blood-pressure and sugar checks.

Group Activities

4. Create a presentation on improving immunization coverage in Punjab. (Model guide)

Model presentation guide: Slides: (1) Why immunization matters: it prevents deadly childhood diseases. (2) Current gaps: missed children in remote areas, myths, and lost vaccination cards. (3) Solutions: door-to-door campaigns, school and mosque announcements, lady health workers' follow-up visits, and record cards kept safely. (4) Call to action: every child fully vaccinated on schedule, with parents reminded of the next due date.

Chapter 2: Gastrointestinal System

A. Select the correct answers for the following questions.

1. The function of the small intestine is:

(a) Absorption of water

(b) Absorption of nutrients ✓

(c) Secretion of bile

(d) Storage of waste

Correct answer: Absorption of nutrients

A. Select the correct answers for the following questions.

2. Which organ produces bile?

(a) Pancreas

(b) Stomach

(c) Liver ✓

(d) Small intestine

Correct answer: Liver

A. Select the correct answers for the following questions.

3. Which diagnostic procedure uses a flexible tube to examine the GI tract?

(a) X-ray

(b) Endoscopy ✓

(c) MRI

(d) CT scan

Correct answer: Endoscopy

A. Select the correct answers for the following questions.

4. What role does the liver play in digestion?

(a) Absorbing nutrients

(b) Producing digestive enzymes

(c) Producing bile ✓

(d) Breaking down proteins

Correct answer: Producing bile

A. Select the correct answers for the following questions.

5. What is the main function of the large intestine?

(a) Digest proteins

(b) Absorb water and electrolytes ✓

(c) Break down fats

(d) Produce digestive enzymes

Correct answer: Absorb water and electrolytes

A. Select the correct answers for the following questions.

6. Which pathogen causes typhoid fever?

(a) Vibrio cholerae

(b) Salmonella Typhi ✓

(c) E. coli

(d) Entamoeba histolytica

Correct answer: Salmonella Typhi

A. Select the correct answers for the following questions.

7. Which of the following is a risk factor for GERD?

(a) Eating large meals before bed ✓

(b) Excessive fibre intake

(c) Drinking water frequently

(d) High intake of protein

Correct answer: Eating large meals before bed

B. Write short answers.

1. Define peptic ulcer and GERD.

A peptic ulcer is a sore in the lining of the stomach or the first part of the small intestine (duodenum), caused mainly by stomach acid, infection with Helicobacter pylori, or overuse of painkillers like NSAIDs. GERD (gastro-oesophageal reflux disease) is the repeated flowing back of stomach acid into the oesophagus, causing heartburn and irritation of its lining.

B. Write short answers.

2. What is the function of the villi in the small intestine?

Villi are tiny finger-like projections lining the small intestine. They hugely increase its inner surface area, so digested nutrients can be absorbed efficiently into the blood. Each villus also contains tiny blood vessels that carry the absorbed nutrients away to the body.

B. Write short answers.

3. How is the typhoid fever transmitted?

Typhoid fever is transmitted by the faecal-oral route: the bacterium Salmonella Typhi spreads through food or water contaminated with the faeces of an infected person or carrier. Poor handwashing, unsafe water and unhygienic food handling are the usual routes.

B. Write short answers.

4. What are the common symptoms of peptic ulcers?

The common symptoms are a burning or gnawing pain in the upper abdomen, especially when hungry or at night, bloating, nausea or vomiting, loss of appetite, and feeling full quickly. Severe cases may show vomiting of blood or black stools, which need urgent medical care.

B. Write short answers.

5. What are some preventive measures for cholera?

Drink only purified, boiled or bottled water; wash hands thoroughly with soap before eating and after using the toilet; store food properly in a refrigerator and avoid undercooked food; eat a balanced diet and stay hydrated; and use vaccination against cholera where it is offered. These are the chapter's own preventive measures for GI infections.

C. Write answers in detail.

1. Discuss the functions of major organs of the GI system.

The mouth chews food and starts starch digestion with saliva. The oesophagus carries food to the stomach, which churns it and digests proteins with acid and enzymes. The small intestine completes digestion and absorbs nutrients through its villi. The large intestine absorbs water and electrolytes and stores waste. The liver makes bile, which helps digest fats; the gallbladder stores it; the pancreas supplies digestive enzymes and insulin; and the rectum and anus expel waste.

C. Write answers in detail.

2. Describe the causes, symptoms, and treatments for GERD and peptic ulcer.

GERD is caused by a weak valve between oesophagus and stomach, large meals before bed, obesity, spicy or fatty food and smoking; its symptoms are heartburn, acid taste, chest discomfort and cough; treatment is smaller meals, avoiding triggers, not lying down after eating, and acid-reducing medicines. Peptic ulcer is caused mainly by H. pylori infection, NSAID overuse and excess acid; symptoms are burning upper-abdominal pain, bloating and nausea; treatment is antibiotics for H. pylori, acid-reducing medicines, avoiding NSAIDs and alcohol, and medical follow-up.

C. Write answers in detail.

3. Write pathogens, transmission, symptoms and complications of cholera, typhoid fever, and gastroenteritis.

Cholera: pathogen Vibrio cholerae; transmission through contaminated water/food; symptoms are sudden watery diarrhoea, vomiting and dehydration; complications are severe dehydration, shock and death if untreated. Typhoid: pathogen Salmonella Typhi; faecal-oral transmission through contaminated food/water; symptoms are prolonged fever, headache, abdominal pain and weakness; complications include intestinal bleeding or perforation. Gastroenteritis: pathogens include viruses, bacteria and parasites; transmission through contaminated food, water or contact; symptoms are diarrhoea, vomiting, cramps and fever; complications are dehydration, especially dangerous in children.

C. Write answers in detail.

4. Discuss the role of diagnostic tools (e.g., blood tests, imaging, endoscopy) in diagnosing GI disorders.

Blood tests: a Complete Blood Count (CBC) can show infection (raised white cells), and Liver Function Tests (LFTs) measure enzymes and proteins to assess the liver and detect hepatitis. Imaging: X-rays outline the stomach or intestines, CT scans give detailed abdominal images for tumours, inflammation or blockages, MRI gives detailed soft-tissue images, and ultrasound examines the liver, gallbladder and pancreas. Endoscopy uses a flexible tube with a camera (endoscope) to look directly inside the GI tract and diagnose its disorders.

Group Activities

1. Write a reflective note on the importance of early diagnosis in managing GI diseases. (Model answer)

Model note: Early diagnosis matters because GI diseases like ulcers, hepatitis and cancers are far easier to treat when found early, before complications such as bleeding, severe dehydration or organ damage develop. Simple steps, such as reporting persistent stomach pain, diarrhoea or vomiting promptly, using blood tests, imaging and endoscopy when advised, and completing treatment, save lives and reduce cost and suffering for families.

Practical Activities

1. Identify and label parts of GI system on models/diagrams. (Guide)

Guide: On a GI model or diagram, label in order: mouth, oesophagus, stomach, liver, gallbladder, pancreas, small intestine, large intestine (colon), rectum and anus. Point to the liver for bile production, the stomach for protein digestion, and the small intestine with its villi for nutrient absorption, and trace the path of food from mouth to anus.

Chapter 3: Respiratory System

A. Select the correct answers for the following questions.

1. Which part of the respiratory system is responsible for gas exchange?

(a) Trachea

(b) Bronchi

(c) Alveoli

(d) Pharynx

Correct answer: Alveoli

A. Select the correct answers for the following questions.

2. What is the function of the diaphragm?

(a) To filter the air

(b) To exchange gases

(c) To control the expansion and contraction of the lungs

(d) To produce mucus

Correct answer: To control the expansion and contraction of the lungs

A. Select the correct answers for the following questions.

3. How does pneumonia affect the lungs?

(a) It fills the alveoli with fluid

(b) It removes mucus from the airways

(c) It increases oxygen levels in the blood

(d) It strengthens the lung tissues

Correct answer: It fills the alveoli with fluid

A. Select the correct answers for the following questions.

4. Which respiratory infection can spread through contaminated surfaces and direct contact?

(a) Tuberculosis

(b) COVID-19 ✓

(c) Asthma

(d) Pneumonia

Correct answer: COVID-19

A. Select the correct answers for the following questions.

5. What is the mode of transmission for influenza?

(a) Mosquito bites

(b) Direct contact with contaminated blood

(c) Respiratory droplets from coughs and sneezes ✓

(d) Insect bites

Correct answer: Respiratory droplets from coughs and sneezes

A. Select the correct answers for the following questions.

6. What is a common symptom of tuberculosis?

(a) Short-term headache

(b) Persistent cough lasting more than three weeks ✓

(c) Skin rash

(d) Increased appetite

Correct answer: Persistent cough lasting more than three weeks

A. Select the correct answers for the following questions.

7. Which of the following is the best way to prevent the spread of COVID-19?

(a) Taking antibiotics regularly

(b) Wearing masks and practicing social distancing ✓

(c) Avoiding all fruits and vegetables

(d) Sleeping for more than 12 hours a day

Correct answer: Wearing masks and practicing social distancing

A. Select the correct answers for the following questions.

8. Which of the following can be diagnosed using pulmonary function tests?

(a) Diabetes

(b) Asthma and COPD ✓

(c) Liver cirrhosis

(d) High blood pressure

Correct answer: Asthma and COPD

A. Select the correct answers for the following questions.

9. Which imaging technique uses strong magnetic fields and radio waves to create detailed images of soft tissues in the lungs?

(a) CT scan

(b) MRI ✓

(c) PET scan

(d) X-ray

Correct answer: MRI

A. Select the correct answers for the following questions.

10. Which condition may require oxygen therapy?

(a) Anaemia

(b) Chronic respiratory diseases ✓

(c) High blood sugar

(d) Kidney stones

Correct answer: Chronic respiratory diseases

B. Write short answers.

1. How does the process of gas exchange take place in the alveoli?

Alveoli are tiny air sacs with very thin walls wrapped in capillaries. Oxygen from inhaled air diffuses across the alveolar wall into the blood, while carbon dioxide from the blood diffuses into the alveoli to be breathed out. This exchange works because gases move from higher to lower concentration across the thin membrane.

B. Write short answers.

2. Discuss the symptoms and causes of asthma.

Asthma symptoms are wheezing, coughing (often at night), shortness of breath and a feeling of tightness in the chest. It is caused by inflamed, sensitive airways that narrow in response to triggers such as dust, pollen, smoke, cold air, exercise or respiratory infections.

B. Write short answers.

3. What is the significance of vaccination in preventing respiratory infections like influenza and COVID-19?

Vaccination trains the immune system to recognize the virus, so infection is less likely and, if it happens, illness is usually milder with fewer complications and hospitalizations. The chapter notes the flu vaccine reduces severe illness, and COVID-19 vaccination, masks and distancing together reduce spread. Vaccination also protects vulnerable people who cannot be vaccinated.

B. Write short answers.

4. Identify the pathophysiology and clinical presentation of chronic obstructive pulmonary disease (COPD).

In COPD the airways and alveoli are progressively damaged, usually by long-term smoking or air pollution, so airflow is limited and the lungs cannot

empty properly. Its clinical presentation is chronic cough with sputum, increasing breathlessness on exertion, wheezing and frequent chest infections, worsening slowly over years.

B. Write short answers.

5. Write the names of the diagnostic methods used to evaluate lung function and respiratory conditions.

Pulmonary Function Tests (PFTs, including spirometry) measure how much air can be breathed in and out and how well the lungs work; chest X-ray and CT/MRI imaging show lung structure; pulse oximetry measures blood oxygen; sputum and blood tests detect infection; and a stethoscope examination identifies abnormal breathing sounds like wheezing or crackling.

C. Write answers in detail.

1. Discuss the function of the respiratory system.

The respiratory system brings oxygen into the body and removes carbon dioxide: air enters through nose and mouth, passes the pharynx, trachea and bronchi into the lungs, and gas exchange occurs in the alveoli. It also warms, moistens and filters inhaled air, supports speech through the voice box, and helps regulate blood acidity by controlling carbon dioxide levels. The diaphragm and chest muscles drive breathing by expanding and contracting the lungs.

C. Write answers in detail.

2. Explain the transmission, symptoms, prevention and treatment for TB, influenza and COVID-19.

TB: caused by Mycobacterium tuberculosis, spread through airborne droplets from coughing; symptoms are persistent cough over three weeks, fever, night sweats and weight loss; prevention is BCG vaccination, ventilation and masks;

treatment is a long course of anti-TB medicines taken completely. Influenza: spread by respiratory droplets; symptoms are fever, cough, sore throat, body aches and fatigue; prevention is flu vaccination and hygiene; treatment is rest, fluids and medicines for symptoms. COVID-19: spread by droplets, contaminated surfaces and direct contact; symptoms are fever, cough, loss of taste/smell and breathlessness; prevention is vaccination, masks and distancing; treatment is supportive care, with hospital care and oxygen for severe cases.

C. Write answers in detail.

3. Describe the role of pulmonary function tests (PFTs) and imaging techniques in diagnosing respiratory conditions.

Pulmonary Function Tests measure how much air the lungs can hold and how fast it can be breathed out, which shows whether airflow is blocked (as in asthma and COPD) and how severe it is. Imaging adds structure: chest X-rays reveal infections, fluid or masses; CT scans give detailed cross-sections for tumours or chronic damage; MRI uses magnetic fields and radio waves for detailed soft-tissue images. Together, function tests plus imaging let doctors diagnose, grade and follow respiratory disease.

C. Write answers in detail.

4. Describe the key symptoms and treatment options for pneumonia and how it differs from other respiratory infections.

Pneumonia is infection of the lung tissue itself: its alveoli fill with fluid or pus, causing fever, cough with sputum, chest pain and breathlessness, and low blood oxygen in severe cases. Treatment depends on the cause: antibiotics for bacterial pneumonia, supportive care (rest, fluids, fever control) for viral cases, and hospital oxygen or ventilation for severe illness; the pneumonia vaccine prevents bacterial pneumonia. It differs from other respiratory infections

because it affects the deep lung tissue and gas exchange, whereas bronchitis affects the airways and influenza is a more general viral illness with body aches and fatigue.

Group Activities

1. Identify the major structures of the respiratory system using diagrams/models. (Guide)

Guide: On a diagram or model, identify and label: nose, pharynx, larynx, trachea, bronchi, bronchioles, alveoli, lungs, diaphragm and ribs. Trace air from the nose through the trachea and bronchi to the alveoli, and point to the diaphragm as the muscle that expands and contracts the lungs.

Group Activities

2. Develop a public health poster on preventing respiratory infections like TB and influenza. (Model guide)

Model poster guide: Title: Stop Coughs Spreading. Points: cover coughs and sneezes; wash hands often; wear a mask when ill; keep rooms ventilated; get flu and BCG vaccinations; seek care for a cough lasting more than three weeks, fever or weight loss. A healthy lungs message: avoid smoking and polluted air.

Group Activities

3. Design a community awareness campaign on the dangers of smoking and air pollution, and prevention of respiratory infections (e.g., TB, influenza, COVID-19). (Model guide)

Model campaign guide: Message: smoking and polluted air damage lungs and raise the risk of COPD, lung cancer and infections. Activities: school assembly talk, posters at clinics, a no-smoking pledge, and a clean-air walk. Prevention steps to promote: quit smoking (it helps repair lung damage), avoid smoky

rooms, vaccinate, wash hands, mask when ill, and test promptly for persistent cough.

Group Activities

4. Role-play a patient consultation to explain the purpose of a diagnostic test like a chest X-ray. (Model guide)

Model role-play guide: Doctor: A chest X-ray is a quick, painless picture of your lungs that shows infection, fluid or other problems we cannot see from outside. Patient: Is it safe? Doctor: It uses a very small amount of radiation and takes only a few minutes. It helps us confirm pneumonia or TB and choose the right treatment, rather than guessing.

Practical Activities

1. Nebulization Therapy: Use a nebulizer to give medication to the lungs, helping treat conditions like asthma or COPD. (Guide)

Guide: Under supervision, place the prescribed medicine in the nebulizer cup, attach the mask or mouthpiece, and switch on so a fine mist is formed. The patient breathes the mist in normally until the medicine is finished, delivering the drug directly to the airways to open them and ease breathing in asthma or COPD.

Practical Activities

2. Pulmonary Function Test: Measure how much air you can breathe in and out to check how well your lungs work. (Guide)

Guide: Using a spirometer under supervision, breathe in fully, seal the lips around the mouthpiece, and blow out as hard and as long as possible. The readings show lung capacity and airflow, helping detect asthma or COPD and track whether treatment is working.

Practical Activities

3. Assessment of Respiratory Rate: Count how many breaths a person takes in one minute to check their breathing rate. (Guide)

Guide: Let the person rest and breathe normally without knowing they are being counted. Count each full breath (in and out) for one minute by watching the chest rise, and record the rate; a normal resting adult rate is about 12-20 breaths per minute, and a much faster rate suggests illness.

Practical Activities

4. Use of Stethoscope to Identify Breathing Sounds: Use a stethoscope to listen to lung sounds and identify any problems like wheezing or crackling. (Guide)

Guide: Place the stethoscope chest piece on the chest and back while the person breathes deeply. Normal breathing sounds are soft and even; wheezing is a high whistling sound of narrowed airways (as in asthma), and crackling is a bubbling sound of fluid or mucus in the lungs.

Practical Activities

5. CO_2 and Lime Water Test (Exhaled Air Experiment): Blow air into lime water and see how it changes color, showing the presence of carbon dioxide in exhaled air. (Guide)

Guide: Blow gently through a straw into a test tube of clear lime water for a minute. The lime water turns milky because exhaled carbon dioxide reacts with calcium hydroxide to form white calcium carbonate. This proves exhaled air contains much more carbon dioxide than inhaled air.

Practical Activities

6. Use of Pulse Oximeter to Measure Oxygen Saturation: Use a pulse oximeter to check how much oxygen is in a person's blood. (Guide)

Guide: Clip the pulse oximeter on a fingertip, keep the hand still, and read the oxygen saturation (SpO2) and pulse rate. A normal SpO2 is about 95-100%; a lower reading, especially with breathlessness, needs prompt medical attention.

Chapter 4: Circulatory System

A. Select the correct answers for the following questions.

1. What is the main function of the circulatory system?

(a) To filter waste products

(b) To transport oxygen and nutrients ✓

(c) To regulate body temperature

(d) To digest food

Correct answer: To transport oxygen and nutrients

A. Select the correct answers for the following questions.

2. Which blood vessels carry blood away from the heart to the other parts of the body?

(a) Veins

(b) Arteries ✓

(c) Capillaries

(d) Lymphatic vessels

Correct answer: Arteries

A. Select the correct answers for the following questions.

3. What is the term for high blood pressure?

(a) Coronary artery disease

(b) Heart failure

(c) Hypertension ✓

(d) Angina

Correct answer: Hypertension

A. Select the correct answers for the following questions.

4. Which of the following is a risk factor for coronary artery disease?

(a) Regular exercise

(b) High cholesterol ✓

(c) Low blood pressure

(d) Healthy diet

Correct answer: High cholesterol

A. Select the correct answers for the following questions.

5. Which diagnostic test measures the electrical activity of the heart?

(a) Angiogram

(b) Stress test

(c) ECG (Electrocardiogram) ✓

(d) X-ray

Correct answer: ECG (Electrocardiogram)

A. Select the correct answers for the following questions.

6. What is a common symptom of heart failure?

(a) Chest pain

(b) Shortness of breath ✓

(c) Nausea

(d) Increased appetite

Correct answer: Shortness of breath

A. Select the correct answers for the following questions.

7. Which lifestyle change can help prevent heart disease?

(a) Smoking

(b) Eating a high-fat diet

(c) Regular physical activity ✓

(d) Excessive alcohol consumption

Correct answer: Regular physical activity

A. Select the correct answers for the following questions.

8. An angiogram helps in diagnosing heart conditions by:

(a) Using sound waves to check heart function

(b) Monitoring electrical signals of the heart

(c) Detecting the presence of blockages in blood vessels ✓

(d) Measuring blood pressure over time

Correct answer: Detecting the presence of blockages in blood vessels

A. Select the correct answers for the following questions.

9. What is a major complication of untreated hypertension?

(a) Heart attack ✓

(b) Low blood sugar

(c) Excessive sweating

(d) Weight loss

Correct answer: Heart attack

B. Write short answers.

1. What is the function of capillaries in the circulatory system?

Capillaries are the tiniest blood vessels, with walls only one cell thick. They link arteries to veins and are the place where exchange happens: oxygen and nutrients pass from blood into tissues, while carbon dioxide and wastes pass from tissues into blood.

B. Write short answers.

2. Write the risk factors for hypertension.

The risk factors are: high salt intake, obesity, lack of physical activity, smoking,

excessive alcohol, chronic stress, older age, and a family history of high blood pressure. Several of these can be controlled by lifestyle change.

B. Write short answers.

3. Define coronary artery disease and its primary cause.

Coronary artery disease is the narrowing or blockage of the arteries that supply the heart muscle itself. Its primary cause is atherosclerosis: cholesterol and fatty deposits (plaque) build up on the artery walls, reducing blood flow to the heart.

B. Write short answers.

4. What is an ECG, and what does it measure?

An ECG (electrocardiogram) is a test that records the electrical activity of the heart through electrodes placed on the skin. It measures heart rate, rhythm and the timing of electrical signals, helping detect heart attacks, rhythm problems and strain on the heart.

C. Write answers in detail.

1. Describe the function of the heart.

The heart is a muscular pump with four chambers: the right side pumps blood to the lungs to pick up oxygen, and the left side pumps oxygen-rich blood to the whole body. Its valves keep blood flowing one way, and its steady beating supplies every tissue with oxygen and nutrients while carrying away carbon dioxide and wastes.

C. Write answers in detail.

2. Write note on the causes, symptoms, and treatment for hypertension.

Causes: high salt intake, obesity, inactivity, smoking, alcohol, stress and family history; often no single cause is found. Symptoms: hypertension usually has no

clear symptoms, which is why it is called a silent condition, but severe cases may cause headache, dizziness or nosebleeds. Treatment: less salt, healthy weight, regular exercise, no smoking, limited alcohol, and blood-pressure medicines taken regularly, with periodic checks.

C. Write answers in detail.

3. Discuss the causes, symptoms, and treatment options for heart failure.

Heart failure means the heart cannot pump blood well enough for the body's needs. Causes include coronary artery disease, long-standing hypertension, heart attack damage and faulty valves. Symptoms are shortness of breath, fatigue, swollen ankles and reduced ability to exercise. Treatment is medicines to help the heart and remove excess fluid, lifestyle changes (low salt, activity as advised, no smoking), treatment of the underlying cause, and in advanced cases devices or surgery.

C. Write answers in detail.

4. Describe the diagnostic tests used to assess heart function and how they help in diagnosing cardiovascular conditions.

ECG records the heart's electrical activity and detects rhythm problems and heart attacks. An angiogram injects dye into the coronary arteries and takes X-ray pictures to detect blockages. A stress test records the heart while exercising to reveal problems hidden at rest. Echocardiography uses sound waves to show the heart's chambers, valves and pumping strength, and blood-pressure measurement plus blood tests complete the assessment. Together they show whether the heart's structure, rhythm and blood supply are normal.

Group Activities

1. Make a presentation on the impact of lifestyle choices on cardiovascular health and preventive measures. (Model guide)

Model presentation guide: Slides: (1) How smoking, fatty salty food, inactivity and alcohol damage the heart. (2) Healthy choices from the chapter's figure: cardiovascular exercise, sufficient sleep, yoga and meditation, mindful eating, monitoring heart rate, staying hydrated, managing chronic diseases, and quitting smoking and alcohol. (3) Preventive checks: blood pressure, cholesterol and sugar. (4) Message: small daily choices protect the heart for life.

Group Activities

2. Identify the major components of the circulatory system using diagrams or models. (Guide)

Guide: On a diagram or model, label: heart (four chambers), arteries (carry blood away from the heart), veins (return blood to the heart), capillaries (exchange vessels), and blood itself. Trace the double circulation: heart to lungs and back, then heart to body and back.

Group Activities

3. Design a community awareness campaign on the importance of heart health. (Model guide)

Model campaign guide: Message: heart disease is largely preventable. Activities: a walk for heart health, free blood-pressure checking camp, posters on salt, smoking and exercise, and a talk by a doctor. Slogan: Check your pressure, move daily, eat mindfully, and never smoke.

Practical Activities

1. Role-play a patient consultation to explain the purpose of a diagnostic test like an ECG. (Model guide)

Model role-play guide: Doctor: An ECG is a painless test that records your heart's electrical activity through small stickers on your chest. It shows whether your heartbeat is regular and whether the heart is under strain. Patient: Does it hurt? Doctor: No, it takes only a few minutes and helps us decide your treatment.

Practical Activities

2. Heart Rate Measurement: Measure the number of heart beats per minute during rest and after exercise. (Guide)

Guide: Find the pulse at the wrist, count beats for 15 seconds and multiply by 4 to get beats per minute at rest. Repeat immediately after exercise and compare: the rate rises with activity and should settle back toward resting within a few minutes in a healthy person.

Practical Activities

3. Pulse Measurement: Check the pulse at different points on the body to evaluate heart rate. (Guide)

Guide: Feel the pulse at the wrist (radial), neck (carotid) and, with care, other points, counting beats per minute at each. The rates should agree; a very fast, slow or irregular pulse, or a pulse that differs between points, should be reported to a teacher or doctor.

Practical Activities

4. Use of Sphygmomanometer: Use a sphygmomanometer to measure blood pressure. (Guide)

Guide: Wrap the cuff around the upper arm at heart level, inflate it, then slowly release while listening with a stethoscope (or reading the digital display). The first sound/reading is systolic pressure and the last is diastolic, written for example as 120/80 mmHg; high readings need rechecking and medical advice.

Practical Activities

5. Use of Stethoscope for Heart Sounds: Use a stethoscope to listen to heart sounds. (Guide)

Guide: Place the stethoscope on the chest over the heart and listen while the person rests quietly. Normal sounds are the paired lub-dub of valves closing; extra sounds, murmurs or an irregular rhythm are abnormal findings to report, not to diagnose alone.

Practical Activities

6. Use of Pulse Oximeter: Use a pulse oximeter to measure blood oxygen levels. (Guide)

Guide: Clip the pulse oximeter on a fingertip, keep the hand still, and read blood oxygen saturation (SpO2) and pulse rate. Normal SpO2 is about 95-100%; low readings with breathlessness or chest symptoms need prompt medical attention.

Chapter 5: Urinary System

A. Select the correct answers for the following questions.

1. The kidneys play a role in maintaining homeostasis. They regulate:

(a) The levels of glucose in the blood

(b) The concentration of waste products, water, and salts ✓

(c) The production of white blood cells in response to infections

(d) The breakdown of fats for energy

Correct answer: The concentration of waste products, water, and salts

A. Select the correct answers for the following questions.

2. Which bacteria is the most common cause of UTIs?

(a) Streptococcus pneumoniae

(b) Escherichia coli (E. coli) ✓

(c) Mycobacterium tuberculosis

(d) Salmonella typhi

Correct answer: Escherichia coli (E. coli)

A. Select the correct answers for the following questions.

3. Which of these is a UTI of kidneys?

(a) Urethritis

(b) Cystitis

(c) Pyelonephritis ✓

(d) Nephritis

Correct answer: Pyelonephritis

A. Select the correct answers for the following questions.

4. What is the purpose of lithotripsy in kidney stone treatment?

(a) Breaks stones ✓

(b) Removes stones surgically

(c) Dissolves stones with drugs

(d) Prevents new stones

Correct answer: Breaks stones

A. Select the correct answers for the following questions.

5. Which lab test helps to diagnose Chronic Kidney Disease (CKD)?

(a) Blood glucose test

(b) Serum creatinine test ✓

(c) Liver function test

(d) Haemoglobin test

Correct answer: Serum creatinine test

A. Select the correct answers for the following questions.

6. What is the main characteristic of Acute Kidney Injury (AKI)?

(a) Sudden kidney damage ✓

(b) Gradual kidney failure

(c) Permanent kidney loss

(d) Infection-related damage

Correct answer: Sudden kidney damage

A. Select the correct answers for the following questions.

7. Which of the following is a key difference between AKI and CKD?

(a) AKI is slow, CKD is sudden

(b) AKI is permanent, CKD is reversible

(c) AKI is reversible, CKD is progressive ✓

(d) AKI occurs only in elderly people

Correct answer: AKI is reversible, CKD is progressive

A. Select the correct answers for the following questions.

8. Which lab result is most commonly elevated in both AKI and CKD?

(a) Blood glucose

(b) Serum creatinine ✓

(c) Platelet count

(d) Haemoglobin level

Correct answer: Serum creatinine

A. Select the correct answers for the following questions.

9. Which blood test is most useful for assessing kidney function?

(a) Haemoglobin test

(b) Liver enzyme test

(c) Serum creatinine test ✓

(d) Cholesterol test

Correct answer: Serum creatinine test

A. Select the correct answers for the following questions.

10. What is the normal range of serum creatinine in healthy adults?

(a) 0.1-0.3 mg/dL

(b) 0.5-1.2 mg/dL ✓

(c) 2.0-3.5 mg/dL

(d) 4.0-5.5 mg/dL

Correct answer: 0.5-1.2 mg/dL

B. Write short answers.

1. What is the main function of the urinary system?

The urinary system filters the blood in the kidneys, removes waste products and excess water and salts as urine, and thereby keeps the body's internal balance

(homeostasis). It also helps control blood pressure and the body's water and salt levels.

B. Write short answers.

2. Name two common symptoms of a urinary tract infection (UTI).

Two common symptoms are a burning feeling while passing urine and a frequent, urgent need to urinate with only small amounts passed. Other symptoms can be cloudy or bloody urine, lower abdominal pain and fever.

B. Write short answers.

3. What is the most common cause of kidney stones?

Kidney stones most commonly form when urine becomes concentrated with minerals, especially calcium oxalate, which crystallize instead of staying dissolved. Low fluid intake, high salt or oxalate intake and repeated urinary infections increase this risk.

B. Write short answers.

4. What is the key difference between Acute Kidney Injury (AKI) and Chronic Kidney Disease (CKD)?

AKI is sudden kidney damage, often over hours or days, that is frequently reversible if the cause is treated promptly. CKD is a gradual, progressive loss of kidney function over months or years, usually from diabetes or hypertension, which is generally not reversible.

B. Write short answers.

5. What is pyelonephritis, and how does it occur?

Pyelonephritis is a urinary tract infection of the kidneys. It usually occurs when bacteria, most often E. coli, travel upward from the bladder through the ureters

into the kidney, causing fever, flank pain, nausea and painful urination.

B. Write short answers.

6. Why is urinalysis important in detecting urinary infections?

Urinalysis examines urine for pus cells, bacteria, blood, protein and glucose. Finding bacteria and white cells points to infection, while blood or protein can reveal kidney damage, so the test detects urinary infection and kidney disease early, cheaply and non-invasively.

C. Write answers in detail.

1. Explain how the urinary system helps maintain homeostasis in the body.

The kidneys filter blood and adjust what leaves in urine: they excrete excess water, salts (sodium, potassium) and wastes like urea and creatinine, while keeping needed substances. By doing this they stabilize blood volume and pressure, salt balance, acid-base balance (pH) and waste levels. When water is scarce they conserve it (concentrated urine); when there is excess they excrete it, keeping the internal environment steady despite changing intake.

C. Write answers in detail.

2. What are kidney stones? Write a note on the treatments for kidney stones.

Kidney stones are hard deposits of minerals and salts, usually calcium oxalate, that form inside the kidney when urine is too concentrated. Small stones may pass with plenty of fluids and pain relief. Larger stones are treated by lithotripsy, which uses shock waves to break the stone into small pieces that can pass out, or by surgical removal when needed. Prevention is drinking enough water, limiting excess salt and oxalate-rich foods, and treating infections promptly.

C. Write answers in detail.

3. Describe the transmission, symptoms, and prevention of pyelonephritis, cystitis, and urethritis.

These are UTIs at different levels, usually caused by bacteria (most often E. coli) entering through the urethra. Urethritis is infection of the urethra, causing burning urination and discharge. Cystitis is bladder infection, causing frequent painful urination, lower abdominal pain and cloudy urine. Pyelonephritis is kidney infection, with fever, flank/back pain, nausea and vomiting in addition. Prevention for all three: drink enough water, urinate regularly and after intercourse, wipe front to back, maintain personal hygiene, and treat lower UTIs promptly before they ascend to the kidneys.

C. Write answers in detail.

4. What is urinalysis? What abnormalities in urine indicate diseases?

Urinalysis is the laboratory examination of a urine sample: its colour, clarity, pH, and the presence of cells and chemicals. Abnormalities and their meaning: pus cells and bacteria indicate urinary infection; red blood cells indicate bleeding from the urinary tract or kidney disease; protein indicates kidney damage; glucose suggests diabetes; crystals suggest stone-forming risk; and dark or very concentrated urine can reflect dehydration. Abnormal results are confirmed with further tests.

C. Write answers in detail.

5. Explain the significance of blood tests like serum creatinine and GFR in diagnosing kidney function.

Serum creatinine is a waste product from muscles that healthy kidneys excrete; when kidney function falls, its blood level rises (normal about 0.5-1.2 mg/dL in adults), so a raised value signals kidney dysfunction in both AKI and CKD. GFR (glomerular filtration rate) is calculated from creatinine, age, sex and body size

and estimates how much blood the kidneys filter per minute; a falling GFR stages CKD severity and guides treatment. Blood Urea Nitrogen (BUN) adds information: normal 7-20 mg/dL, above 20 suggesting dysfunction or dehydration, and above 50 severe impairment.

Group Activities

1. Label a diagram of the urinary system. (Guide)

Guide: On a urinary system diagram, label: two kidneys, ureters (tubes carrying urine down), urinary bladder (stores urine), and urethra (carries urine out). Point to the kidney as the filter, and trace urine from kidney through ureter to bladder to urethra.

Group Activities

2. Create a flowchart showing the progression and management of chronic kidney disease. (Model guide)

Model flowchart: Risk factors (diabetes, hypertension) lead to early kidney damage (protein in urine, rising creatinine) which progresses through CKD stages as GFR falls, to kidney failure if unmanaged. Management at each stage: control blood sugar and pressure, low-salt healthy diet, avoid kidney-harming medicines, regular creatinine/GFR checks, and in failure, dialysis or transplant assessment.

Group Activities

3. Develop a public health poster on preventing UTIs. (Model guide)

Model poster guide: Title: Stop UTIs Before They Start. Points: drink plenty of clean water; do not hold urine for long; wash hands and maintain personal hygiene; wipe front to back; urinate after intercourse; wear clean cotton underwear; and seek care promptly for burning urination or fever instead of

waiting.

Group Activities

4. Role-play a patient consultation to explain the purpose of a diagnostic test like a kidney ultrasound. (Model guide)

Model role-play guide: Doctor: A kidney ultrasound is a painless scan using sound waves to make pictures of your kidneys. It shows their size, stones, blockages or swelling without radiation. Patient: Will it hurt? Doctor: No, it takes a few minutes and helps us find the cause of your pain or abnormal blood test.

Practical Activities

1. Develop a model to demonstrate the process of kidney filtration. (Guide)

Guide: Build a model with a funnel and filter paper standing for the kidney's filter: pour a mixture of water with dissolved salt (standing for wastes) and beads (standing for blood cells) through it. The filtrate passing through stands for urine forming, while the beads held back stand for blood cells kept in the blood, showing selective filtration.

Practical Activities

2. Perform an experiment to observe salt crystallization as a simulation of kidney stone formation. (Guide)

Guide: Dissolve as much salt as possible in warm water to make a saturated solution, pour it into a dish, and leave it to evaporate slowly. As water leaves, salt crystals grow, simulating how concentrated urine lets minerals crystallize into kidney stones. Note: drinking enough water keeps urine dilute and reduces this risk.

Chapter 6: First Aid-I

A. Select the correct answers for the following questions.

1. What does the "D" in DRABC stand for?

(a) Diagnosis

(b) Danger

(c) Delivery

(d) Duration

Correct answer: Danger

A. Select the correct answers for the following questions. (FLAGGED ITEM - see answer)

2. Which of the following is NOT a sign of cardiac arrest? Options: (a) Sudden collapse (b) Chest pain (c) Bluish skin (d) Dizziness. NOTE: No option is marked correct - see answer.

FLAGGED - NO CORRECT OPTION AS PRINTED: The chapter's own table (book p. 65, section 6.2.1 Recognizing Signs of Cardiac Arrest) lists ALL FOUR options as signs or symptoms of cardiac arrest: sudden collapse, chest pain, bluish skin (cyanosis) and dizziness/lightheadedness all appear in that table. The question asks for one option that is NOT a sign, but no such option exists among the four printed choices, so none is marked correct. The chapter's listed signs are: sudden collapse, unresponsiveness, no pulse, no breathing, bluish skin and dilated pupils.

A. Select the correct answers for the following questions.

3. What is the first step in CPR for an adult?

(a) Start chest compressions

(b) Call emergency services

(c) Open the airway

(d) Provide rescue breaths

Correct answer: Call emergency services

A. Select the correct answers for the following questions.

4. What is the correct compression depth for CPR on an adult?

(a) 1 inch

(b) 2 inches

(c) 3 inches

(d) 4 inches

Correct answer: 2 inches

A. Select the correct answers for the following questions.

5. Which type of wound is caused by friction?

(a) Laceration

(b) Puncture

(c) Abrasion

(d) Incision

Correct answer: Abrasion

A. Select the correct answers for the following questions.

6. Which method is used for a suspected muscle or tendon injury?

(a) R.I.C.E

(b) A.B.C

(c) D.R.A

(d) I.C.E

Correct answer: R.I.C.E

A. Select the correct answers for the following questions.

7. Which of these agencies provides first aid services in Punjab?

(a) Rescue 1122

(b) Edhi Foundation

(c) Pakistan Red Crescent

(d) All of the above

Correct answer: All of the above

A. Select the correct answers for the following questions.

8. What should you do if you find a person with a suspected spinal fracture?

(a) Move them to a safe place

(b) Immobilize the person and call for help

(c) Apply cold therapy

(d) Encourage movement

Correct answer: Immobilize the person and call for help

B. Write short answers.

1. What does the DRABC approach stand for in first aid?

DRABC stands for: D - Danger (check the scene is safe), R - Response (check whether the person responds), A - Airway (check and clear the airway), B - Breathing (look, listen and feel for normal breathing), C - Circulation (check for a pulse and signs of circulation; if absent, start chest compressions).

B. Write short answers.

2. What are the signs of cardiac arrest?

According to the chapter's table, a person in cardiac arrest may show: sudden collapse, unresponsiveness, no pulse or heartbeat, no breathing, bluish skin (cyanosis) and dilated pupils. Associated symptoms listed are chest pain, dizziness or lightheadedness, weak or irregular pulse and sudden fatigue.

B. Write short answers.

3. How should you perform chest compressions during CPR?

Place the heel of the hand on the centre of the chest, push down about 2 inches (5-6 cm) at a rate of 100-120 compressions per minute, and let the chest fully recoil between pushes. For an adult, give 30 compressions followed by 2 rescue breaths, and continue without long pauses until help arrives or the person recovers.

B. Write short answers.

4. What is the primary first aid for a person suffering from burns?

Cool the burn at once with cool running water for at least 10-20 minutes, remove jewellery or tight items near the burn before swelling starts, and cover it loosely with a sterile non-stick dressing. Do not pop blisters and do not apply toothpaste, butter or ice directly; for large, deep or facial burns, seek emergency medical help immediately.

B. Write short answers.

5. What is the R.I.C.E method for sprain management?

R.I.C.E stands for: Rest - avoid using the injured area to prevent further damage; Ice - apply an ice pack wrapped in cloth for 15-20 minutes every 2-3 hours to reduce swelling and pain; Compression - wrap the injured area with an elastic bandage; Elevation - raise the injured limb above heart level to reduce swelling.

C. Write answers in detail.

1. Explain the DRABC approach in first aid and its significance in saving lives.

DRABC is the primary survey done before any treatment: Danger - assess hazards and make the area safe, approaching safely and calling for help if it is not safe; Response - talk to or tap the person gently to check responsiveness,

and call for help if needed; Airway - assess, tilt the head and lift the chin, and clear obstructions; Breathing - look, listen and feel for normal breathing, since without oxygen permanent brain damage occurs within minutes; Circulation - check respiration and pulse, and if the person is unresponsive and not breathing regularly, call for help and start CPR if trained. Its significance is that it finds and fixes life threats in order, within the golden hour (the first 60 minutes after injury, most crucial for survival), instead of treating minor injuries first.

C. Write answers in detail.

2. Describe the steps and techniques involved in performing CPR on an adult.

Steps from the chapter: ensure the scene is safe and no hazards are present; tap the person and shout to check responsiveness; call emergency services (for example Rescue 1122 in Pakistan) for help; open the airway by tilting the head and lifting the chin; check breathing; if there is no normal breathing, start chest compressions in the centre of the chest, about 2 inches deep, at 100-120 per minute, allowing full recoil; give 2 rescue breaths after every 30 compressions; and continue CPR with minimal pauses until emergency help arrives, an AED is available, or the person starts breathing normally.

C. Write answers in detail.

3. Discuss the different types of wounds and their appropriate first aid management.

Abrasion is a scrape from friction: clean it gently with water, remove dirt, and cover with a clean dressing. Laceration is a torn, irregular wound: apply firm pressure to stop bleeding, clean around it, and dress it; deep or gaping lacerations need medical care and possibly stitches. Puncture is a deep narrow wound from a sharp object: do not remove a deeply embedded object, control bleeding around it, and seek medical help because of infection and tetanus risk.

Incision is a clean cut: apply pressure, clean, close the edges with a dressing, and seek care if it is deep, long or still bleeding.

C. Write answers in detail.

4. Explain the differences between fractures, sprains, and strains. Discuss their first aid management.

A fracture is a broken bone: signs are pain, swelling, deformity and inability to move the part; first aid is to immobilize the part (splint it as found), do not move the person unnecessarily, and call for help - especially for a suspected spinal fracture, where the person must be immobilized and not moved. A sprain is injury to a ligament at a joint, and a strain is injury to a muscle or tendon; both show pain, swelling and limited movement and are managed by R.I.C.E (Rest, Ice, Compression, Elevation) and medical assessment if severe. The key difference: fractures involve bone and need immobilization and hospital care, while sprains and strains involve soft tissue and respond to R.I.C.E.

Group Activities

1. Your school will manage the demonstration of first aid by Rescue 1122. After attending these demonstrations, role play for demonstrating first aid in different emergency situations. (Model guide)

Model guide: After the Rescue 1122 demonstration, practise in groups: (1) a collapsed person - DRABC check, call 1122, start CPR on a mannequin if trained; (2) a bleeding wound - gloves on, pressure, dressing; (3) a sprain - R.I.C.E steps; (4) a burn - cool, cover, do not apply home remedies. Each group explains what it did and why, and the teacher corrects technique.

Group Activities

2. Create a response plan for emergencies at school, home or public places.

(Model guide)

Model response plan: (1) Stay calm and check danger. (2) Call Rescue 1122 and give the exact location and what happened. (3) Give first aid within your training (DRABC, bleeding control, R.I.C.E). (4) Send someone to guide the ambulance in and to inform parents/teachers. (5) Keep a first-aid kit, emergency numbers and exit routes known to everyone, and practise the plan with drills.

Group Activities

3. Simulate the ability to assess an emergency scene for safety. (Model guide)

Model guide: Walk up to a mock scene and name hazards before touching the patient: traffic, fire, electricity, broken glass, crowd, or an unstable structure. Decide: is it safe to approach? If not, stay back and call for help. Only when safe, approach, introduce yourself, and begin the DRABC primary survey.

Practical Activities

1. Perform chest compressions and rescue breaths under supervision on a mannequin. (Guide)

Guide: Under a trained supervisor and on a mannequin only, practise: scene safety, responsiveness check, call for help, airway opening, 30 chest compressions (2 inches deep, 100-120 per minute) followed by 2 rescue breaths, repeated in cycles. Never practise compressions on a real person.

Practical Activities

2. Simulate bleeding wound dressing using fake wounds or red-coloured liquid. (Guide)

Guide: Using a fake wound and red-coloured liquid, practise: wear gloves, apply firm pressure with a clean cloth, place a sterile dressing, bandage firmly but not so tight that circulation stops, and raise the limb. Practise explaining when to call

for help: heavy, spurting or unstoppable bleeding.

Practical Activities

3. Illustrate proper burn treatment, including cooling, covering, and avoiding incorrect methods. (Guide)

Guide: Demonstrate on a model: cool the burn with running water, cover loosely with a sterile dressing, and list incorrect methods to avoid: ice directly on skin, toothpaste, butter, popping blisters, and removing stuck clothing. State clearly when to call Rescue 1122: large, deep, facial, hand or genital burns, or burns on a child.

Chapter 7: Physical Education

A. Select the correct answers for the following questions.

1. What is the primary purpose of Physical Education?

(a) To entertain students

(b) To promote physical fitness and health ✓

(c) To teach academic subjects

(d) To prepare students for competitive sports

Correct answer: To promote physical fitness and health

A. Select the correct answers for the following questions.

2. Which of the following is a benefit of regular exercise?

(a) Increased stress and anxiety

(b) Improved cardiovascular health ✓

(c) Decreased muscle strength

(d) Poor posture

Correct answer: Improved cardiovascular health

A. Select the correct answers for the following questions.

3. What does the "I" in the RICE method stand for?

(a) Ice ✓

(b) Injury

(c) Intensity

(d) Ignition

Correct answer: Ice

A. Select the correct answers for the following questions.

4. Which exercise primarily strengthens the core?

(a) Squats

(b) Push-ups

(c) Plank ✓

(d) Walking lunges

Correct answer: Plank

A. Select the correct answers for the following questions.

5. What does overexertion during physical activity increase the risk of?

(a) Better performance

(b) Heart attack and muscle strain ✓

(c) Enhanced endurance

(d) Improved flexibility

Correct answer: Heart attack and muscle strain

A. Select the correct answers for the following questions.

6. What is one of the benefits of strength training?

(a) Decreases muscle size

(b) Increases endurance ✓

(c) Reduces bone density

(d) Decreases flexibility

Correct answer: Increases endurance

A. Select the correct answers for the following questions.

7. Which of the following can improve metabolic health?

(a) Regular exercise ✓

(b) Poor diet

(c) Lack of sleep

(d) Sedentary lifestyle

Correct answer: Regular exercise

A. Select the correct answers for the following questions.

8. Why is maintaining proper posture important during exercises like squats?

(a) To increase speed

(b) To prevent strain on joints and muscles ✓

(c) To reduce flexibility

(d) To improve muscle hypertrophy

Correct answer: To prevent strain on joints and muscles

A. Select the correct answers for the following questions.

9. What can poor posture during exercise lead to?

(a) Reduced risk of injuries

(b) Muscle imbalances and joint damage ✓

(c) Faster recovery

(d) Improved heart health

Correct answer: Muscle imbalances and joint damage

A. Select the correct answers for the following questions.

10. Which of these exercises is recommended to improve lower body strength and balance?

(a) Push-ups

(b) Walking lunges ✓

(c) Bicycle crunches

(d) Mountain climbers

Correct answer: Walking lunges

B. Write Short Answer

1. What are the mental and emotional benefits of regular exercise?

Regular exercise reduces stress and anxiety, improves mood through

endorphins, builds confidence and self-esteem, and improves sleep and concentration. It also gives a healthy outlet for emotions and a sense of achievement, which supports overall mental well-being.

B. Write Short Answer

2. List the risks associated with overexertion during physical activity.

Overexertion risks include muscle strain and joint injury, exhaustion and dehydration, dizziness or fainting, and in extreme cases heart attack, especially in unfit people or in heat. It also causes prolonged soreness and can force a long break from activity, undoing fitness gains.

B. Write Short Answer

3. How does exercise contribute to weight management?

Exercise burns calories, builds muscle that raises resting energy use, and improves metabolism and appetite control. Combined with a balanced diet, it helps prevent weight gain and supports steady, healthy fat loss without losing muscle.

B. Write Short Answer

4. What are the dangers of poor posture during exercise?

Poor posture during exercise puts uneven strain on joints and muscles, causing muscle imbalances, joint damage, back and knee pain, and a higher risk of sprains, strains and long-term injury. It also reduces the effectiveness of the exercise, since the wrong muscles do the work.

B. Write Short Answer

5. What are the benefits of muscle strength training?

Strength training increases muscle size and endurance, strengthens bones,

improves posture, balance and metabolism, and makes daily tasks easier. It also protects joints, helps weight management, and reduces the risk of falls and injuries.

C. Write answer in detail:

1. Discuss the physical, mental, and emotional benefits of regular exercise. How do these benefits contribute to an individual's overall well-being?

Physical benefits: stronger heart and lungs, better cardiovascular health, stronger muscles and bones, improved flexibility, balance and metabolic health, and healthier weight. Mental benefits: sharper focus, better sleep and lower stress and anxiety. Emotional benefits: better mood, confidence and social connection through team activity. Together they form overall well-being: a fit body supports a calm mind, and a calm mind supports healthy habits, so the person functions better at school, work and home.

C. Write answer in detail:

2. Explain the process of how exercise strengthens the heart and improves cardiovascular health

During exercise the heart must pump more blood, so it works harder and, over weeks, becomes stronger and more efficient: it pumps more blood per beat, so resting heart rate falls. Blood vessels become more flexible, blood pressure and cholesterol improve, and the body uses oxygen better. The result is improved cardiovascular health: lower risk of heart disease, hypertension and stroke, and greater stamina for daily activity.

C. Write answer in detail:

3. What is the significance of the RICE method in managing sports injuries? Provide a detailed explanation of each step involved.

RICE matters because quick, correct first aid limits swelling, pain and further damage after a sports injury, speeding recovery. Rest: stop using the injured part at once to prevent worse damage. Ice: apply an ice pack wrapped in cloth for 15-20 minutes every 2-3 hours to reduce swelling and pain (never ice directly on skin). Compression: wrap with an elastic bandage, firm but not circulation-stopping, to control swelling. Elevation: raise the injured limb above heart level so fluid drains away and swelling falls. If pain, swelling or inability to move persists, seek medical care.

C. Write answer in detail:

4. What are the consequences of poor posture and improper techniques during exercise? How can these be avoided?

Consequences are muscle imbalances, joint damage, back, knee and shoulder pain, strains and sprains, and slower progress because the wrong muscles work. They can be avoided by: learning correct technique from a teacher or coach before adding weight or speed; warming up and stretching; keeping the back straight, knees aligned and core engaged (for example in squats); using suitable footwear and equipment; progressing gradually; and stopping when pain, not just effort, appears.

Group Activities

1. Write a short essay on the importance of physical education in daily life. (Model answer)

Model essay: Physical education is important in daily life because it builds the fitness, strength and habits that keep the body and mind healthy. Regular activity prevents obesity, heart disease and diabetes, improves mood, sleep and concentration, and teaches discipline, teamwork and fair play. In a life of long sitting and screens, daily physical education is the simplest protection for

lifelong health.

Group Activities

2. Create a poster highlighting the benefits of regular exercise. (Model guide)

Model poster guide: Title: Move Daily, Live Healthy. Points: stronger heart and muscles; better mood and sleep; healthy weight; stronger bones; less stress. Pictures: walking, running, cycling, sports. Footer line: at least 30 minutes of activity most days, with warm-up and correct posture.

Group Activities

3. Perform a simulation of first aid for a sports injury. (Model guide)

Model simulation guide: Scenario: a player twists an ankle. Steps to demonstrate: stop play, check danger and response, keep the player still, apply R.I.C.E (rest, wrapped ice 15-20 minutes, elastic compression, elevation), and arrange medical assessment if the player cannot bear weight or swelling is severe. Narrate each step as it is done.

Group Activities

4. Conduct a group discussion on the importance of exercise for heart health. (Model guide)

Model discussion guide: Questions: How does exercise strengthen the heart? What happens to heart rate and blood pressure with regular activity? Which activities suit our age group? What stops people exercising, and how can we help each other stay active? Conclusion to record: regular moderate activity, correct posture and gradual progress protect the heart, while overexertion and poor technique cause harm.

Group Activities

5. Create a presentation on the role of exercise in combating obesity. (Model guide)

Model presentation guide: Slides: (1) What obesity is and why it harms health. (2) How exercise burns calories, builds muscle and improves metabolism. (3) A practical weekly plan: daily walking, sports, strength work twice a week, less screen sitting. (4) Diet plus activity together: balanced meals, less sugary drinks and junk food. (5) Message: small daily steps, kept up, beat short intense efforts.

Practical Activities

1. Measure and record the heart rate and respiratory rate (before and after exercise). (Guide)

Guide: At rest, count the wrist pulse for 15 seconds and multiply by 4 for beats per minute, and count breaths for one minute. Repeat immediately after exercise and record both sets. Compare: both rates rise with exercise and should return toward resting values within a few minutes; a very slow recovery should be reported.

Practical Activities

2. Demonstrate basic first aid techniques for treating a sprain or strain. (Guide)

Guide: Demonstrate R.I.C.E on a partner or model: rest the part, apply wrapped ice for 15-20 minutes, apply an elastic bandage, and elevate the limb. Explain what not to do: do not massage forcefully, do not apply heat at first, and do not continue playing through severe pain.

Practical Activities

3. Conduct the common posture mistakes and correction activity. (Guide)

Guide: Show common mistakes: rounded back in squats, knees collapsing inward, slouched sitting, and head jutting forward. For each, demonstrate the

correction: straight back with chest up, knees tracking over toes, shoulders back, and core engaged. Practise in pairs, with one partner observing and correcting the other.

Chapter 8: Nutrition

A. Select the correct answers for the following questions.

1. Why are fats essential despite being high in calories?

(a) Store vitamins and energy ✓

(b) Main energy source

(c) Aid protein digestion

(d) Regulate water balance

Correct answer: Store vitamins and energy

A. Select the correct answers for the following questions.

2. Why are proteins important?

(a) Main energy source

(b) Build and repair tissues ✓

(c) Store energy

(d) Needed only for muscles

Correct answer: Build and repair tissues

A. Select the correct answers for the following questions.

3. Pale skin and fatigue suggest a deficiency of?

(a) Vitamin A

(b) Calcium

(c) Iron ✓

(d) Protein

Correct answer: Iron

A. Select the correct answers for the following questions.

4. Which food group helps in muscle growth and repair?

(a) Vegetables

(b) Fruits

(c) Meat, pulses, and eggs ✓

(d) Oil and fat

Correct answer: Meat, pulses, and eggs

A. Select the correct answers for the following questions.

5. Why should oil and fats be consumed in moderation?

(a) They contain no nutrients

(b) They can cause muscle loss

(c) Excess intake leads to obesity and heart diseases ✓

(d) They reduce calcium absorption

Correct answer: Excess intake leads to obesity and heart diseases

A. Select the correct answers for the following questions.

6. Which dietary habit increases the risk of diabetes?

(a) Eating a balanced diet

(b) Drinking plenty of water

(c) Consuming too much sugar and refined carbohydrates ✓

(d) Eating high-fibre foods

Correct answer: Consuming too much sugar and refined carbohydrates

A. Select the correct answers for the following questions.

7. What is the main cause of stunting in children?

(a) Lack of sleep

(b) Chronic malnutrition ✓

(c) Excessive sugar intake

(d) Too much physical activity

Correct answer: Chronic malnutrition

A. Select the correct answers for the following questions.

8. Why is anaemia more common in young children and women?

(a) Due to high iron requirements and low intake ✓

(b) Due to excessive protein consumption

(c) Because they eat too many carbohydrates

(d) Because they drink too much water

Correct answer: Due to high iron requirements and low intake

A. Select the correct answers for the following questions.

9. Which of the following is a key sign of wasting?

(a) Excess fat accumulation

(b) Severe weight loss and muscle thinning ✓

(c) High blood pressure

(d) Increased appetite

Correct answer: Severe weight loss and muscle thinning

A. Select the correct answers for the following questions.

10. What is a major health risk associated with obesity?

(a) Improved digestion

(b) Stronger immune system

(c) Increased risk of heart disease and diabetes ✓

(d) Higher bone density

Correct answer: Increased risk of heart disease and diabetes

B. Write short answers.

1. Differentiate between macronutrients and micronutrients with examples.

Macronutrients are needed in large amounts and give energy or structure: carbohydrates (bread, rice), proteins (meat, pulses, eggs) and fats (oil, butter).

Micronutrients are needed in tiny amounts but are essential for body functions: vitamins (for example vitamin A, B, C, D) and minerals (for example iron, calcium, iodine). Both groups are necessary; neither replaces the other.

B. Write short answers.

2. What is the role of proteins in the body?

Proteins build and repair tissues, including muscles, skin and organs, and they make enzymes, hormones and antibodies. They are also needed for growth in children and for healing after injury or illness.

B. Write short answers.

3. Why are vitamins and minerals essential for maintaining good health?

Vitamins and minerals run countless body processes: they support immunity, blood formation, bone strength, vision, nerve function and energy release from food. Even small deficiencies cause disease - for example iron deficiency causes anaemia (pale skin, fatigue), and vitamin B deficiency is worsened by alcohol abuse, which interferes with vitamin B absorption.

B. Write short answers.

4. Define balanced diet. What are the main groups of food in a balanced diet?

A balanced diet is one that provides all nutrients - carbohydrates, proteins, fats, vitamins, minerals and water - in the right amounts for age, sex and activity. Its main food groups are: cereals and grains; meat, pulses and eggs (protein foods); vegetables and fruits; milk and dairy; and oils and fats in moderation, plus plenty of clean water.

B. Write short answers.

5. Differentiate between wasting and stunting.

Wasting is low weight for height, caused by recent severe undernutrition or illness, and its key sign is severe weight loss with muscle thinning. Stunting is low height for age, caused by chronic malnutrition over a long period, especially in early childhood, and it reflects long-term growth failure rather than sudden weight loss.

C. Write answers in detail.

1. Describe the role of carbohydrates, proteins, and fats in the body. Also write their sources.

Carbohydrates are the body's main energy source: sources are bread, rice, potatoes and other cereals. Proteins build and repair tissues and make enzymes and antibodies: sources are meat, fish, pulses, eggs and milk. Fats store energy, store fat-soluble vitamins, insulate the body and protect organs: sources are oil, butter, ghee, nuts and fatty fish. All three are needed in balance; excess fat in particular leads to obesity and heart disease, so oils and fats are taken in moderation.

C. Write answers in detail.

2. Explain the importance of a balanced diet and how dietary guidelines help in preventing malnutrition and chronic diseases.

A balanced diet gives every nutrient in the right amount, supporting growth, immunity, energy and repair, and preventing both deficiency diseases and excess-weight diseases. Dietary guidelines help by translating this into daily practice: eat varied foods from all groups, favour whole grains, vegetables, fruits and pulses, limit sugar, salt and excess fat, drink clean water, and adjust portions to age and activity. Following them prevents malnutrition (undernutrition and micronutrient deficiency) and lowers the risk of chronic diseases like diabetes, heart disease and obesity.

C. Write answers in detail.

3. Define malnutrition and differentiate between undernutrition and overnutrition. Provide examples and their consequences.

Malnutrition is poor nutrition from too little, too much, or an imbalanced intake of nutrients. Undernutrition is too little food or nutrients: examples are wasting, stunting and anaemia, with consequences of weak immunity, poor growth and learning, and higher death risk in children. Overnutrition is excess intake, especially of energy, sugar and fat: examples are overweight and obesity, with consequences of diabetes, heart disease, joint problems and reduced life expectancy. Both forms can exist in the same community or even household.

C. Write answers in detail.

4. Write notes on stunting, wasting, and obesity. Explain their causes, symptoms, consequences, and preventive measures.

Stunting: cause is chronic malnutrition in early childhood; symptoms are short height for age and delayed development; consequences are poor learning and lifelong reduced productivity; prevention is adequate maternal and child nutrition, breastfeeding, and clean water and sanitation. Wasting: cause is recent severe food shortage or illness; symptoms are severe weight loss and muscle thinning; consequences are weak immunity and high death risk; prevention is prompt treatment of illness, therapeutic feeding and food security. Obesity: cause is excess energy intake with low activity; symptoms are excess body fat and breathlessness on exertion; consequences are diabetes, heart disease and joint disease; prevention is balanced diet, portion control, regular exercise and less sugary drinks and junk food.

C. Write answers in detail.

5. What is anaemia? Explain its causes, symptoms, effects on the body, and how

it can be prevented.

Anaemia is a condition in which the blood has too few red blood cells or too little haemoglobin, so it carries less oxygen. Causes: iron deficiency (the commonest, from low iron intake or blood loss), vitamin B12 or folate deficiency, chronic disease, and, in children and women, high iron requirements with low intake. Symptoms: pale skin, fatigue, weakness, dizziness and breathlessness. Effects: poor growth and learning in children, reduced work capacity, and in pregnancy, risk to mother and baby. Prevention: eat iron-rich foods (meat, pulses, green vegetables) with vitamin C foods to aid absorption, take iron supplements where advised (and vitamin B-rich foods or supplements if needed), treat blood loss and infections, and avoid alcohol abuse, which interferes with vitamin B absorption.

Group Activities

1. Design a balanced meal plan for a day based on dietary guidelines. (Model answer)

Model meal plan: Breakfast: whole-grain bread or paratha with egg, plus milk and a fruit. Lunch: rice or chapati with pulses or meat, vegetables and salad, plus yoghurt. Snack: fruit or nuts, with clean water through the day. Dinner: chapati with vegetables and pulses or fish, plus salad. Principles: all food groups included, oils and sugar kept moderate, portions matched to activity.

Group Activities

2. Create a chart listing examples of macronutrients and micronutrients and their food sources. (Model answer)

Model chart: Macronutrients - Carbohydrates: bread, rice, potatoes; Proteins: meat, pulses, eggs, milk; Fats: oil, butter, nuts. Micronutrients - Vitamins: vitamin A (carrots, liver), B vitamins (whole grains, meat, fish, eggs), C (citrus, guava);

Minerals: iron (meat, pulses, green vegetables), calcium (milk, yoghurt), iodine (iodized salt).

Group Activities

3. Write a short essay on the importance of following dietary guidelines. (Model answer)

Model essay: Dietary guidelines matter because they turn nutrition science into daily choices anyone can follow. Eating varied foods from all groups, limiting sugar, salt and fat, and drinking clean water prevents both undernutrition and chronic disease. A family that follows them spends less on illness, children grow and learn better, and adults stay active longer. Guidelines are not a diet for a week; they are a lifelong pattern of eating well.

Group Activities

4. Create a poster highlighting the effects of malnutrition on health. (Model guide)

Model poster guide: Title: Malnutrition Harms Every Age. Panels: Child stunting (short, slow learning); Wasting (thin, weak, often ill); Anaemia (pale, tired, breathless); Obesity (diabetes, heart disease). Bottom line: eat a balanced diet from all food groups, and seek care early for weight loss, paleness or poor growth.

Group Activities

5. Write a report on the prevention and management of iron deficiency anaemia. (Model answer)

Model report: Iron deficiency anaemia is commonest in young children and women because iron needs are high while intake is low. Prevention: iron-rich foods (meat, pulses, green vegetables) eaten with vitamin C foods, iron

supplements where a doctor advises, treatment of blood loss and worms, and food fortification. Management: confirm with a blood test (haemoglobin), take prescribed iron regularly, recheck blood levels, and follow up diet and causes so it does not return.

Group Activities

6. Create a presentation on the role of nutrition in preventing Type 2 Diabetes. (Model guide)

Model presentation guide: Slides: (1) What Type 2 diabetes is: high blood sugar from insulin resistance, linked to excess weight and diet. (2) Risky habits: too much sugar and refined carbohydrates, sugary drinks, large portions, inactivity. (3) Protective habits: whole grains, vegetables, pulses, fibre, moderate portions, regular activity, healthy weight. (4) Message: daily food choices are the strongest prevention available.

Practical Activities

1. Analyse food samples to detect the presence of macronutrients (starch, proteins, and fats) using standard biochemical tests. (Guide)

Guide: Starch test: add iodine solution to the sample; a blue-black colour shows starch. Protein test (Biuret): add sodium hydroxide then copper sulphate; a violet colour shows protein. Fat test: rub the sample on brown paper; a translucent grease spot that remains shows fat. Record each result in a table and identify which macronutrient each food mainly provides.

Practical Activities

2. Determine the pH levels of different digestive juices and explain their role in digestion. (Guide)

Guide: Using pH paper or a meter under supervision, test samples standing for

digestive juices: gastric juice is strongly acidic (about pH 2) so pepsin can digest protein and germs are killed; saliva is near neutral (about pH 6-7) for starch digestion in the mouth; pancreatic and intestinal juices are alkaline (about pH 8) to neutralize stomach acid so intestinal enzymes can work. Record each pH and its digestive role.